

Specifications – Version 1.0.0

1. Description

The EthnoMK application is a platform where users can upload and browse various costumes, handicrafts, jewelry, textiles, decorations, and traditional items from Macedonian folk and cultural heritage. Users will be able to upload, search, and view images and information about traditional items, organized by region and other criteria.

2. User Requirements

2.1. Functional Requirements

2.1.1. Administrator Module

- 2.1.1.1. The system must provide an option for administrators to delete created records.
- 2.1.1.2. The system must provide an option for administrators to edit created records.

2.1.2. User Module

- 2.1.2.1. The system must allow the user to create a record with an image and content.
- 2.1.2.2. The system must allow the user to list previously created records.
- 2.1.2.3. The system must allow the user to search records by keywords present in existing records.
- 2.1.2.4. The system must provide an option to upload images when creating a new record.
- 2.1.2.5. The system must provide an option to enter data when creating a new record.
- 2.1.2.6. The system must include a section with contact information for technical support.

2.2. Non-Functional Requirements

- 2.2.1. The web application must maintain the same interface regardless of which web browser it is opened in.
- 2.2.2. The web application will operate with its own local database.
- 2.2.3. The application must be scalable.
- 2.2.4. The application must be available 24/7 with minimal downtime.
- 2.2.5. The application will be free to use.
- 2.2.6. The application must be compatible with various devices.
- 2.2.7. The application interface must be easy to use, with a consistent design.

3. Use Case Diagrams

Three use case diagrams are defined for the system:

Fig. 1 – User interaction with the application: Viewing Records

The user can browse all records, open individual records, and extend the view by filtering by

region, filtering by category, or searching by keyword.

Fig. 2 – User interaction with the application: Adding Records

When creating a new record, the user must enter a description/text, attach an image, select a region, and select a category (costumes, jewelry, textiles, handicrafts, etc.). Optionally, the user may also select a material (wool, silk, metal, etc.) and a technique (weaving, embroidery, filigree, etc.). The record is then saved. The user can also access technical support contact.

Fig. 3 – Administrator interaction with the application: Managing Records

The administrator can view records, edit records, delete records, and manage categories/regions (add, delete, update).

4. ER Diagram

The entity-relationship diagram defines the following entities and their relationships:

Entity	Attributes
Record	record_id (PK), title, description, date_created, region_id (FK), category_id (FK), material_id (FK), technique_id (FK), admin_id (FK)
Image	image_id (PK), image_path, record_id (FK)
Admin	admin_id (PK), name, password
Region	region_id (PK), name
Category	category_id (PK), name
Material	material_id (PK), name
Technique	technique_id (PK), name

Relationships: One Admin can manage many Records (1:N). One Record has many Images (1:N). One Region can be associated with many Records (1:N). One Category can be associated with many Records (1:N). One Material can be associated with many Records (1:N). One Technique can be associated with many Records (1:N).

5. UI Design

Two main screens are designed for the application:

Fig. 5 – Home Screen: Features a navigation bar with links to Home, Add Record, and Contact, along with a search bar. The main content area displays a hero text ('Search and discover the rich Macedonian folk heritage') with a keyword search input and a filter dropdown. Below, records are shown as cards in a grid layout, each displaying an image, the item name, a region tag, and a 'View Details' button.

Fig. 6 – Add New Record Screen: A form titled 'Add New Record' with fields for Record Name, Description, Region (checkboxes: Southwest, Polog, Pelagonia, Skopje, Vardar, Northeast, Eastern, Southeast), Category (checkboxes: Costumes, Rugs/Carpets, Carvings, Silver/Metal Items, Wooden Items, Other), Material (checkboxes: Wool, Cotton, Silk, Wood, Metal, Leather, Stone, Other), and Technique (checkboxes: Weaving/Knitting, Embroidery, Engraving/Carving, Drawing/Leatherwork, Printing/Dyeing, Other).